

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1

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Section: B

1. TITLE

Design Static Webpage Using HTML Components

2. AIM

To design a static webpage using pure HTML and semantic HTML components without external CSS.

3. OBJECTIVES

1. To understand the structure of a static HTML webpage.
2. To use semantic HTML elements such as header, nav, main, article, section, aside and footer.
3. To create navigation links and structured webpage content.
4. To create forms using native HTML interaction elements.
5. To understand how browsers render raw HTML using their default styles.

4. THEORY

A static webpage can be created using HTML without external style sheets. Semantic elements provide meaningful structure to the document and help browsers, screen readers and other tools understand the organization of content.

The webpage in this experiment uses a header, navigation area, showcase section, main content area, articles, a sidebar, a form and a footer. Tables are used for structural layout, while native HTML controls provide user interaction.

5. ALGORITHM/PROCEDURE

1. Create a new HTML file.
2. Declare the HTML5 document type and set the language to English.
3. Add the character encoding and webpage title.
4. Create the header with the StaticWeb heading and navigation links.
5. Create the home/showcase section with a heading, description and article link.
6. Create the main content area using a structural table.

7. Add articles explaining core structural tags and native interaction elements.
8. Insert photo elements stacked vertically ("down by down") within the second article section.
9. Create the inquiry form using fieldset, legend, labels, input, select and textarea elements.
10. Create the sidebar using the aside element and display lists of benefits and components.
11. Create the footer with copyright information.
12. Save the file with an .html extension.
13. Open the file in a web browser and verify the output.

6. PROGRAM

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Pure Semantic HTML Static Webpage</title>
</head>
<body>
<!-- Header Section -->
<header>
<table width="100%" border="0" cellpadding="10">
<tr>
<td><h1>StaticWeb</h1></td>
<td align="right">
<nav>
<a href="#home">Home</a> &nbsp; &nbsp; &nbsp;
<a href="#articles">Articles</a>
<a href="#about">About</a> &nbsp; &nbsp; &nbsp;
<a href="#contact">Contact Us</a>
&nbsp; &nbsp; &nbsp; &nbsp;
&nbsp; &nbsp; &nbsp; &nbsp;
</nav>
</td>
</tr>
</table>
</header>
<hr>
<!-- Hero / Showcase Section -->
<section id="home">
<center>
<h2>Semantic Structures Without CSS</h2>
<p>This layout uses raw HTML components to build an organized document hierarchy that browsers can cleanly parse out-of-the-box.</p>
<p><a href="#articles"><b>Read Our Articles Below &ldarr;</b></a></p>
</center>
</section>
<hr>
<!-- Main Content Layout using a Structural Table -->
<table width="100%" border="0" cellpadding="20">
<tr valign="top">
<!-- Left Side: Main Content Column -->
```

```

<td width="70%">
<main id="articles">
<article>
<h2>1. Core Structural Tags</h2>
<p>Published: July 7, 2026</p>
<p>When you omit cascading style sheets entirely, the semantic markup handles all of
the structural heavy lifting. Elements like <main>, <article>, and
<section> tell screen readers and web scrapers exactly how information flows.</p>
<p>Using raw browser defaults ensures excellent loading performance,
perfect accessibility compliance, and absolute structural clarity.</p>
</article>
<br><hr><br>
<article>
<h2>2. Native Interaction Elements</h2>
<p>Published: July 5, 2026</p>
<p>Browsers come equipped with excellent interactive elements by
default. We can capture text inputs, provide multiple-choice configurations, and submit
structured information to backend parsers using pure markup.</p>
<!-- Photos Added Stacked Vertically (Down by Down) -->
<p>
<br><br>

</p>
<h3>Interactive Demonstration Forms</h3>
<form id="contact" action="#" method="post">
<fieldset>
<legend><b>User Inquiry Form</b></legend>
<p>
<label for="name">Full Name: </label>
<input type="text" id="name" name="username" required placeholder="John Doe">
</p>
<p>
<label for="email">Email Address: </label>
<input type="email" id="email" name="useremail" required>
</p>
<p>
<label for="topic">Inquiry Topic: </label>
<select id="topic" name="usertopic">
<option value="general">General Feedback</option>
<option value="technical">Technical Architecture</option>
<option value="academics">Academic Curriculum</option>
</select>
</p>
<p>
<label for="message">Message Details: </label><br>
<textarea id="message" name="usermessage" rows="5" cols="40" placeholder="Type your notes
here..."></textarea>
</p>
<input type="submit" value="Submit Form Data">
<input type="reset" value="Clear Entries">
</fieldset>
</form>
</article>
</main>
</td>
<!-- Right Side: Sidebar Column -->

```

7. CODE EXPLANATION & 8. TEST CASES

Code	Explanation
<!DOCTYPE html>	Declares the document type as HTML5.
<header>	Contains the website heading and navigation links.
<nav>	Provides links to navigate to different sections.
<section>	Groups related content into a logical showcase section.
<main>	Contains the primary body content.
<article>	Contains independent article content.
	Embeds photos/images stacked vertically.
<form>	Provides an interface for entering and submitting data.
<fieldset>/<legend>	Groups related form controls and gives the input block a caption.
<aside>	Contains supplementary sidebar/reference information.
/	Displays unordered and ordered lists.
<footer>	Contains copyright and footer details.

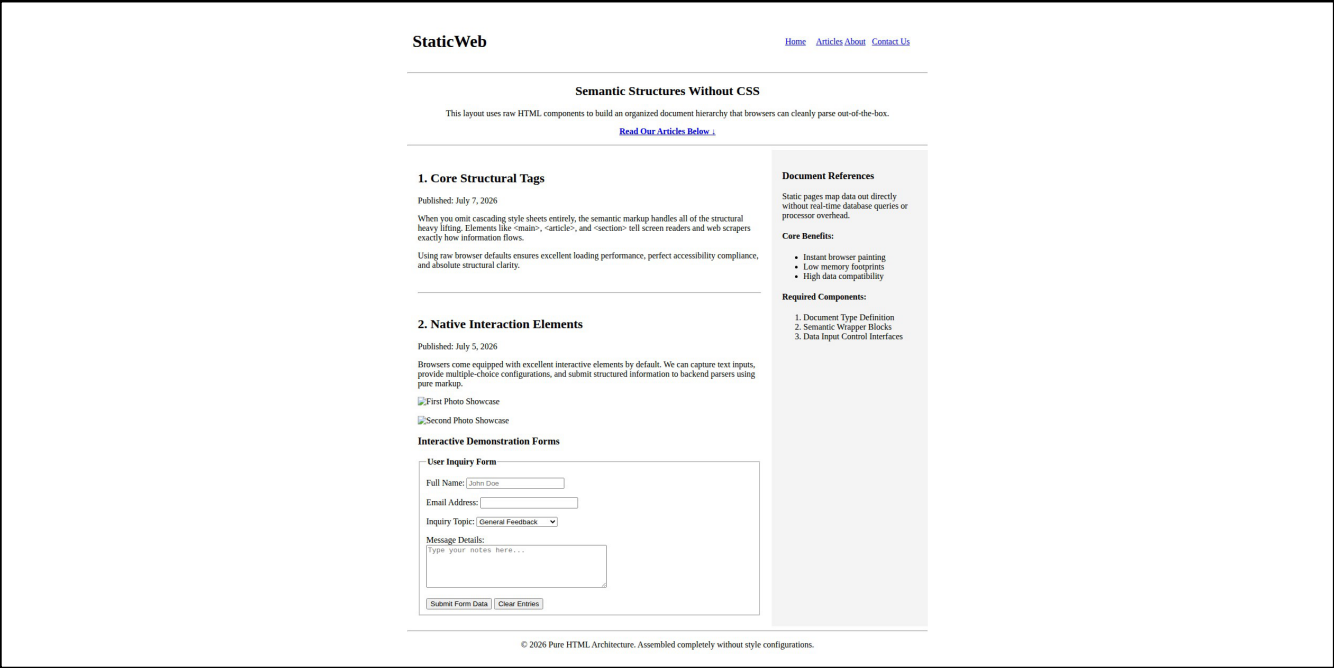
Test Cases and Execution Data

Test Case	Action / Input	Expected Result	Actual Result	Status
TC-01	Load the page in browser	Header, nav, showcase, articles, sidebar and footer render	Displayed correctly	Pass
TC-02	Click the 'Articles' navigation link	Page jumps to the Articles/main content section	Navigated to section correctly	Pass
TC-03	Enter values in Full Name and Email Address	Entered values are accepted and shown in fields	Values accepted correctly	Pass
TC-04	Leave Email Address empty and click Submit Form	Browser shows native required-field validation prompt	Validation prompt displayed as expected	Pass
TC-05	Select 'Technical Architecture' from Dropdown	Dropdown updates to selected option	Selected option displayed correctly	Pass
TC-06	Enter a message and click 'Clear Entries'	All form fields are reset to empty	Fields cleared successfully	Pass

8.1 TEST CASE OUTPUT SCREENSHOTS

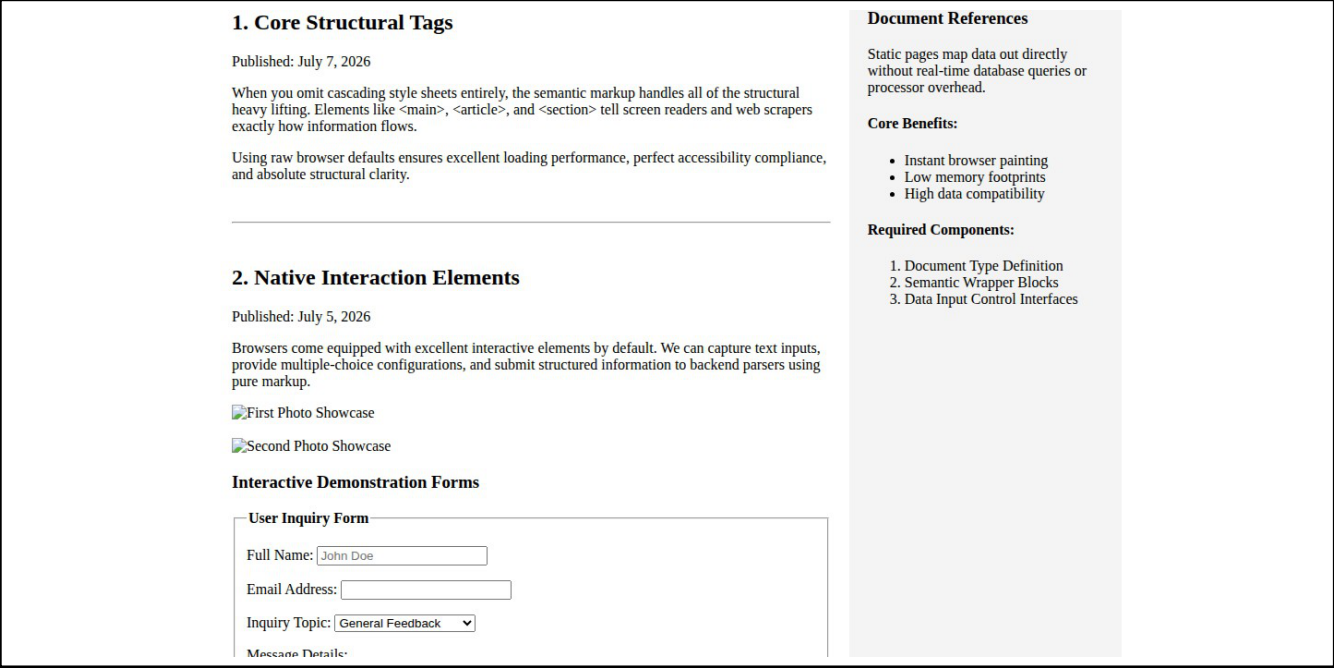
TC-01 Output Screenshot

Load the page in browser



TC-02 Output Screenshot

Click the 'Articles' navigation link



8.1 TEST CASE OUTPUT SCREENSHOTS

TC-03 Output Screenshot

Enter values in the Full Name and Email Address fields

1. Core Structural Tags

Published: July 7, 2026

When you omit cascading style sheets entirely, the semantic markup handles all of the structural heavy lifting. Elements like <main>, <article>, and <section> tell screen readers and web scrapers exactly how information flows.

Using raw browser defaults ensures excellent loading performance, perfect accessibility compliance, and absolute structural clarity.

2. Native Interaction Elements

Published: July 5, 2026

Browsers come equipped with excellent interactive elements by default. We can capture text inputs, provide multiple-choice configurations, and submit structured information to backend parsers using pure markup.

First Photo Showcase

Second Photo Showcase

Interactive Demonstration Forms

User Inquiry Form

Full Name:

Email Address:

Inquiry Topic:

General Feedback

Message Details:

Document References

Static pages map data out directly without real-time database queries or processor overhead.

Core Benefits:

Instant browser painting

Low memory footprints

High data compatibility

Required Components:

1. Document Type Definition

2. Semantic Wrapper Blocks

3. Data Input Control Interfaces

TC-04 Output Screenshot

Leave the Email Address field empty and click 'Submit Form Data'

2. Native Interaction Elements

Published: July 5, 2026

Browsers come equipped with excellent interactive elements by default. We can capture text inputs, provide multiple-choice configurations, and submit structured information to backend parsers using pure markup.

First Photo Showcase

Second Photo Showcase

Interactive Demonstration Forms

User Inquiry Form

Full Name:

Email Address:

Inquiry Topic:

Please fill out this field.

Message Details:

Required Components:

1. Document Type Definition

2. Semantic Wrapper Blocks

3. Data Input Control Interfaces

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8.1 TEST CASE OUTPUT SCREENSHOTS

TC-05 Output Screenshot

Select 'Technical Architecture' from the Inquiry Topic dropdown

2. Native Interaction Elements

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 First Photo Showcase

 Second Photo Showcase

Interactive Demonstration Forms

User Inquiry Form

Full Name:

Email Address:

Inquiry Topic:

Message Details:

Required Components:

1. Document Type Definition
2. Semantic Wrapper Blocks
3. Data Input Control Interfaces

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TC-06 Output Screenshot

Enter a message and click 'Clear Entries'

2. Native Interaction Elements

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 First Photo Showcase

 Second Photo Showcase

Interactive Demonstration Forms

User Inquiry Form

Full Name:

Email Address:

Inquiry Topic:

Message Details:

Required Components:

1. Document Type Definition
2. Semantic Wrapper Blocks
3. Data Input Control Interfaces

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9. OUTPUT

Browser Output (Screenshot):

StaticWeb

[Home](#) | [Articles](#) | [About](#) | [Contact Us](#)

Semantic Structures Without CSS

This layout uses raw HTML components to build an organized document hierarchy that browsers can cleanly parse out-of-the-box.

[Read Our Articles Below](#)

1. Core Structural Tags

Published: July 7, 2026

When you omit cascading style sheets entirely, the semantic markup handles all of the structural heavy lifting. Elements like `<main>`, `<article>`, and `<section>` tell screen readers and web scrapers exactly how information flows.

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Browsers come equipped with excellent interactive elements by default. We can capture text inputs, provide multiple-choice configurations, and submit structured information to backend parsers using pure markup.

Interactive Demonstration Forms

User Inquiry Form

Full Name:

Email Address:

Inquiry Topic:

General Feedback

Message Details:

Type your notes here...

Submit Form Data

Clear Entries

Document References

Static pages map data out directly without real-time database queries or processor overhead.

Core Benefits:

- Instant browser painting
- Low memory footprints
- High data compatibility

Required Components:

- Document Type Definition
- Semantic Wrapper Blocks
- Data Input Control Interfaces

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10. RESULT

Thus, a static webpage was successfully designed using pure HTML and semantic HTML components. The webpage includes navigation, structured articles, a native HTML form, sidebar information, lists and a footer, and can be rendered directly by a web browser without external CSS.